YongHwi Song

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EDUCATION

Mar. 2023 ~ Feb. 2027 (Expected)

Chungbuk National University — Cheongju, Korea
School of Information and Communication Engineering
B.S. in Information and Communication Engineering
GPA: 3.83 / 4.5

RESEARCH INTERESTS

- World Models and World-Action Models
- Model-Based Planning and Reinforcement Learning
- Efficient and Reliable World Models
- World Models for Embodied and Language Agents

RESEARCH EXPERIENCE

- **Artificial Intelligence Laboratory (AI Lab), Chungbuk National University** — Korea
  Research Intern · Advisor: Prof. Keon Myung Lee — Jun. 2026 ~ Present
  Researching efficient and reliable world models and their use in LLM agents.
  - Showed that MPPI planners exploit deterministic rounding error in low-bit TD-MPC2 world models; training-free dithering removed a median 95% of the success drop on Meta-World.
  - Found that latent world-model imagination blurs near misses: success-vs-near-miss AUROC fell from 0.953 to 0.853 on DINO-WM PushT, while far misses stayed separable.
  - Showed that a frozen decision model, JEV, works as an LLM agent's world model without task training, reaching 93.4% success in TextWorld vs. 79.5% for a failure-memory baseline.
- **Multimedia Information Processing (MIP) Laboratory, Chungbuk National University** — Korea
  Research Intern · Advisor: Prof. Hyun Soo Kang — Mar. 2025 ~ Jun. 2026
  Conducted research on multimodal reinforcement learning for autonomous driving.
  - Designed and implemented RCS-AC, a CARLA driving actor-critic that gates RGB, LiDAR, route, and ego-state features by sensor reliability, risk, and Shapley contribution.

ACADEMIC ACTIVITIES & LEADERSHIP

- Runner (13th Cohort) at Hello, World! Season 2 — World Models Paper Review, Pseudo Lab (가짜연구소), Korea (Oct. 2026 ~ Present) / Reviewing world-model research papers, participating in technical discussions, and contributing structured review documents to the project website.

- Team Leader at Applied Machine Learning Division, HyperCore AI & Data Analytics Club, Korea (Jan. 2026 ~ Sep. 2026) / Formed and led a club team for hackathons and paper reviews.

PUBLICATIONS & MANUSCRIPTS

Under Review

- Song, YongHwi* and Lee, Keon Myung
  "Planners Exploit Deterministic Error: Dithered Low-Bit World Models."
  Submitted to the Korean Artificial Intelligence Association (KAIA) Conference (poster), 2026.
  Extended version to be submitted to Robotics: Science and Systems (RSS) 2027.
- Song, YongHwi* and Lee, Keon Myung
  "Near-Miss Blindness: Latent World-Model Imagination Loses Resolution at the Outcome Boundary."
  Submitted to the Korean Artificial Intelligence Association (KAIA) Conference (poster), 2026.
  Extended version to be submitted to Transactions on Machine Learning Research (TMLR).
- Song, YongHwi* and Kang, Hyun Soo
  "Imagine Once, Act Fast: Lightweight Action Readout from a Frozen Driving World-Action Model."
  Submitted to the Korean Artificial Intelligence Association (KAIA) Conference (poster), 2026.

In Preparation

- Song, YongHwi*
  "How Many Candidates Does a One-Step World Model Need? A Random-Codebook Account of Nearest-of-m Training on Frozen DINO Tokens."
  To be submitted to Transactions on Machine Learning Research (TMLR).
- Song, YongHwi*
  "K-PragShift: Diagnosing Korean Pragmatic Revisions in Tool-Using Language Models."
  To be submitted to the EACL 2027 Student Research Workshop.
- Song, YongHwi* and Lee, Keon Myung
  "Imagine Once, Act Fast: Lightweight Action Readout from a Frozen Driving World-Action Model."
  To be submitted to Transactions on Machine Learning Research (TMLR).

Conference Presentations

- Song, YongHwi* and Lee, Keon Myung
  "A World Model for LLM Agents: Comparing a Decision Model with General-Purpose LLMs."
  Oral presentation, 2026 Fall Conference of the Korean Institute of Intelligent Systems (KIIS).

  *First author.

PROJECTS

- Moonkeeper — CLPS Lunar Mission Explorer, 2026 NASA Space Apps Challenge Team Project, Korea / Building a React web app that visualizes CLPS lunar landing sites, Sun/Earth visibility windows, and mission timelines from NASA data. (Sep. 2026 ~ Present)
- Folding — Local-First Document Agent, Team Project, Korea / Built the initial LLM-wiki platform that turns HWP, Office, and PDF files into a linked knowledge base. (Jul. 2026)
- Vehicle and Pedestrian Trajectory Forecasting on Argoverse 2, Personal Project, Korea / Compared linear, LSTM, Transformer, and diffusion trajectory forecasters; PCA latent diffusion reached a minFDE of 0.86 at K=16. (Mar. 2026 ~ Jun. 2026)

COMPETITIONS & HACKATHONS

2026 NASA Space Apps Challenge — Seoul Local Event

- Accepted as a participant in the Seoul Local Event with Team Moonkeeper. (Nov. 2026, upcoming)

2026 Southeast Region Multi-Domain Mobility Business Hackathon — Finalist

- Selected as one of the top 10 teams for the final round. SeaSight / Built a drift world model and drone search planner to find fishers who fall overboard; on 88 real drifter cases, the model cut the search area for 90% containment 2.7×. (Oct. – Nov. 2026)

KRAFTON AI R&D Hackathon — Dream It Yourself, Video-Based Lightweight World-Action Model

- Advanced to the final round. Built a world-action model that predicts cart–double-pendulum dynamics and selects the next action from the same representation. (Oct. 2026)

2026 Startup for All Project (모두의 창업) — Ministry of SMEs and Startups

- Passed the first round. Folding Next / A desktop agent that helps public servants fill forms and draft documents from in-house files, with user-checked sources. (Oct. 2026)

LG Aimers 9th Cohort — KBO Pitch Control Success Probability Prediction AI Hackathon

- Advanced to the Phase 2 online hackathon (top 20%). Worked on predicting pitch-control success probabilities from game context, player history, and tracking data. (Aug. – Sep. 2026)

2026 National Institute of Korean Language AI-Malpyung Competition — Argumentative Writing Scoring

- Advanced to the final round. Developed an AI system for automated evaluation of Korean argumentative writing. (Jul. – Sep. 2026)

2026 CODEGATE AI Startup Hackathon — Finalist

- Selected with Folding for the offline final and demo day (about 20 teams). (Jul. 2026)

3rd MixUp AI Hackathon (Upstage × BDAI) — Finalist

- Solar TutorBoard / Built the agent workflows and backend of a Solar Pro3-based tutoring assistant for lesson reports, payment reminders, and scheduling. (May 2026)

AWARDS AND HONORS

- Academic Excellence Scholarship, Chungbuk National University, Korea (Mar. 2026, Mar. 2024, Sep. 2023)
- Undergraduate Research Assistant Scholarship, Chungbuk National University, Korea (Sep. 2025 ~ Mar. 2026)

ADDITIONAL PUBLICATIONS

1. Hyeban Namgung, Kuk Seorin, Park Geonwoo, Bae Seohyun, Song Yonghwi, Lee Hyesong, Han Jian, Hyojoong Kim, “A Comparative Study on the Prevalence of Forensic Flies Using Chicken Corpse,” Journal of Science Education for the Gifted, 2020.

SKILLS AND TECHNIQUES

- Programming:
  Python, C/C++, Java
- ML Frameworks:
  PyTorch, JAX, Triton, Hugging Face Transformers, scikit-learn, Stable-Baselines3
- World Models & Simulation:
  TD-MPC2, DreamerV3, DINO-WM, MuJoCo, Meta-World, LIBERO, CARLA
- Tools:
  Linux, Git, Hydra, vLLM, FastAPI